

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient performs the task. Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate the performance based on evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

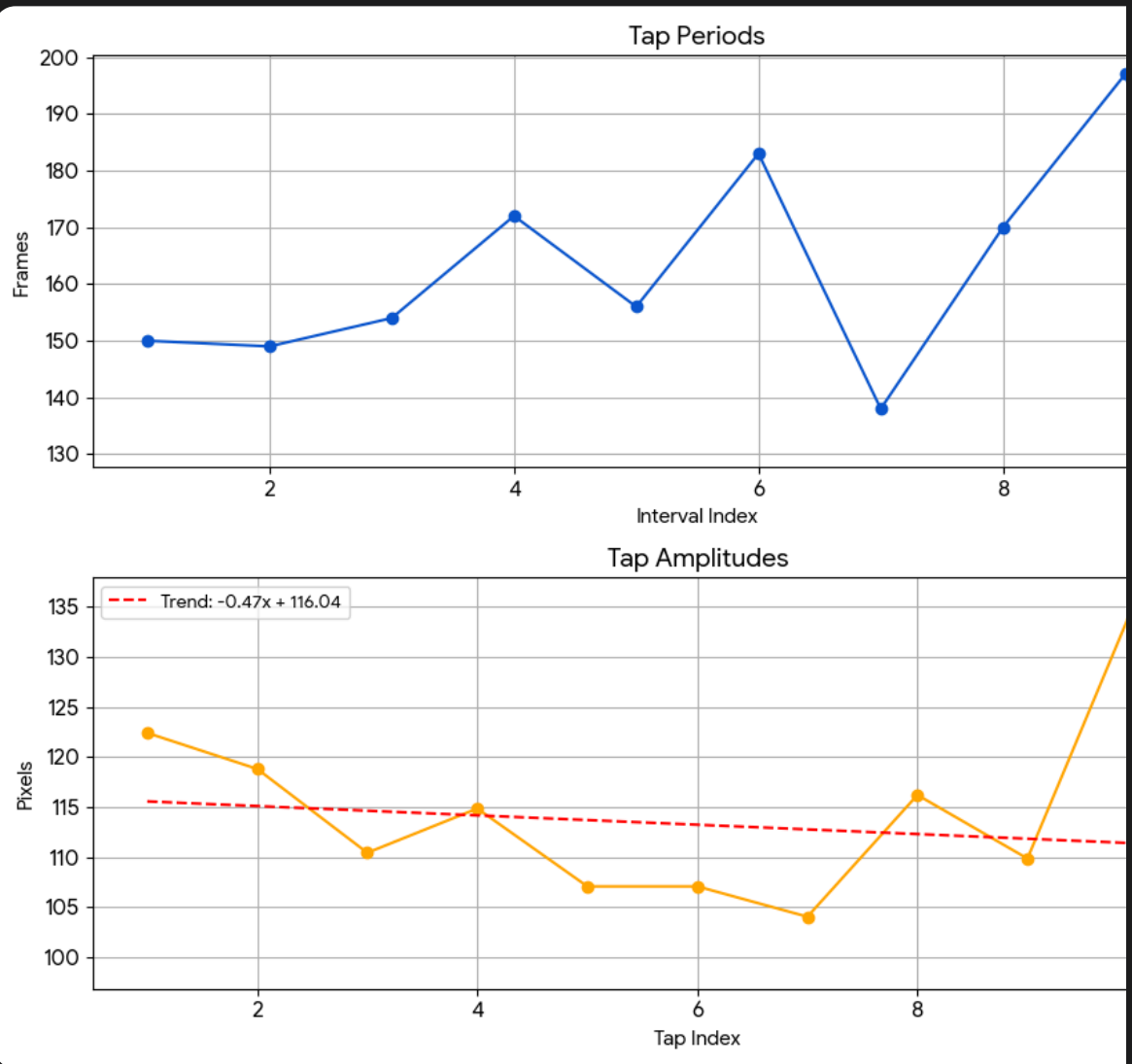
Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide a score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [150, 149, 154, 172, 156, 183, 138, 170, 197, 131]

Tap amplitudes (pixels): [122.41323457861898, 118.81077392223317, 110.43550153822818, 114.856432122881, 107.07474025184464, 107.07474025184464, 104.04326023342406, 116.21101496846157, 109.85900054160, 136.12494260788506, 98.71676655968832]



Based on the extracted features provided, here is the analysis according to the MDS-UPDRS Item 3.4 criteria:

1. Rhythm (Interruptions/Hesitations): The tap periods (in frames) are: [150, 149, 154, 172, 156, 183, 138, 170, 197, 170]

- The baseline rhythm appears to be around 150 frames.
- There are significant deviations indicating hesitations or interruptions at periods **172, 183, 170, and 197**.
- This counts as approximately **4 interruptions** or hesitations.
- *Matches Criteria for Score 2 (Mild): "3 to 5 interruptions during tapping".*

2. Amplitude (Decrement): The tap amplitudes (in pixels) are: [122.4, 118.8, 110.4, 114.9, 107.1, 107.1, 104.0, 136.1, 98.7]

- The amplitude drops from **122.4** (Tap 1) to **118.8** (Tap 2) to **110.4** (Tap 3), and generally trends downward to **104.0** (Tap 7).
- While there is amplitude variability later in the sequence (a rebound at Tap 10), the decrement feature is clearly present **immediately after the 1st tap**.
- *Matches Criteria for Score 3 (Moderate): "The amplitude decrements starting after the 1st tap".*

3. Speed:

- The mean period is 160 frames with high variance (CV ~12%). The irregularity is the most prominent feature rather than progressive slowing.

Conclusion: The subject exhibits signs of **Mild** rhythm disruption (4 interruptions) and **Moderate** amplitude decrement tap). according to the scoring instructions ("Any of the following"), the presence of the criteria for Score 3 (amplitude d early) determines the final score.